

Liveness Tensor

For directly connected datacenter networks

A transitively coherent data structure piggybacking on heartbeat exchanges between nodes. Data is synchronized with a self node at the center of a cluster using progressively smaller data structures. One-hop visibility, useful for cluster algorithms and consensus, provides 4-bit ‘dynamic’ liveness for 8 links. Two-hop visibility for tree healing algorithms, provides ‘stable’ 2-bit liveness for up to 64 cells (32 bytes), and three-hop visibility, for dynamic topology awareness, provides ‘static’ 1 bit topology awareness for up to 512 cells (64 Bytes).

The first use case for the heartbeat architecture is to support tree-less ‘awareness’ of the local topology. Although cells may have a unique identity, they do not have a destination address which (in conventional networks) is conflated with the identity. Instead, cells can address their directly connected (1-hop) neighbors by exchanging messages on the port they connect to their neighbors. This gets us a 1-hop interconnect system. It is possible, but becomes rapidly more difficult to identify specific cells more than one hop away, without some kind of addressing scheme. Our Tree-Based Addressing (TBA) is a name-based addressing scheme that overcomes many of the challenges in today’s networks (particularly fragility to failures, error-prone configuration and security vulnerabilities). Before we get to that next higher ‘voltage’ in our architecture. Let’s first see what we need, and what we can achieve with a small number of hops of transitive connections.

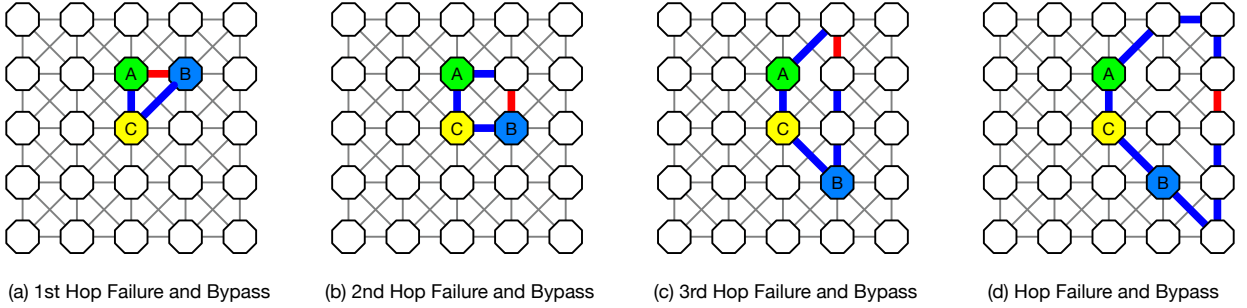


Figure 1: (Liveness-Tensor) Link Failure/Recovery: Local topology awareness from cell (M_0)

Pre-Tree. Coherent awareness of local interconnect topology supports treebuilding, and provides a greater repertoire of algorithms for rapid and deterministic tree healing. Fig. 1 shows four configurations of Alice, Bob and Charlie, where Charlie is the coordinator if his own computational tile. Blue connections are connected, and live, with the heartbeat mechanism. The red links are failed links. Assume Alice and Bob are transferring an atomic token using the 2-phase commit protocol, but the link fails after the first phase and the transaction cannot complete. In (a) Charlie can see that both Alice and Bob have a failed link. As Captain/Coordinator of this transaction, Charlie can retrieve the two halves of the ‘common knowledge’ the left half from Alice, and the right half from Bob, and stitch the two back together (typically by ‘reversing’ the state machine on both sides to bring their state machines back to a quiescent state without losing information). In (b) the situation is slightly more complex because the failure is in the third hop on the left side, and 2nd hop on the right side. This is typical of a ‘failed node’ scenario. In (c) the failed link now shows up in the 3rd hop through Alice, and the 4th hop through Bob. In (d) there can be an arbitrary number of hops between Alice and Bob, and which particular link has failed is not at all obvious. These figures show the progressively decreasing value of ‘local’ awareness, in an mesh interconnect *prior* to the establishment of a routing scheme.

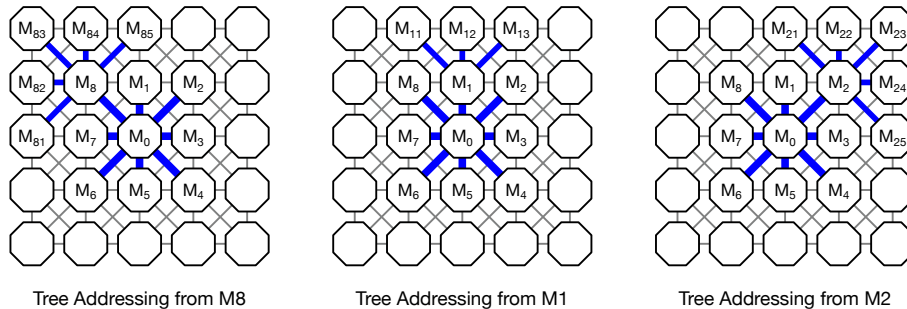


Figure 2: (Relative-Tree-Addressing) Looks different from different from each Tree’s perspective.

Relative Local Addressing

Conventional L2 `src` or `dst` addresses are not needed in the lowest layer (groundplane)¹. Locality (confinement) is provided because destination addresses are implicit in the selected output port, and return addresses are implicit in the same connection. This provides an opportunity for confinement (with significant security benefits), and relative addressing out to only one or two hops. Beyond that Trees need to be stacked.

For the heartbeat architecture, there are three types of slices. That help maintain link (MORT), Cluster, and BRANCH liveness respectively. In the heartbeat architecture, a set of slices is 64 bytes. A slice $s \in S$, is size $S/4$, i.e. 16 Bytes. Typically we send 1, 2 or 3 slices in the heartbeat engine. Slices 2 and 3 are sent *only* if they are needed, and can be pre-empted by the receiving cell, if it wants to begin a transaction as soon as possible². The above figures show 3, and 5 party operations.

Data Structures

The M cluster combines 8 1-hop directly connected MORTS, 8 Ring Bypass MORTS, and 4 2-hop Bypass MORTS. Wherever possible, we try to combine the liveness awareness bits into either 32, or 64 bit words, into 8 or 16 Byte Slices, or 64 Byte data structures that match the physical packet constraints of the packet engine, and Ethernet hardware formats.

For the host superstructure, and carrying data over the PCIe bus (or CXL) it also makes sense to align with 64-bit registers, 64-Byte cache lines, and 4K pages of 64 cache lines. This $64 \times 64 \times 64$ structure may be more than a curious coincidence.

M_0 can directly (from a LOV perspective) address its 8 neighbors, simply by sending packets on that port, and receiving link return packets on the same port.

Hypercell Tiled Architecture

Computation Tiles (1-hop clusters, 9 cell hypercells) are 9 element hardware components (microservers). They form the core architecture. They are based (directly) on [hypergraphs](#) (and their associated hyperlinks)³.

Figures 3 (a, b and c) show hypercell Liveness overlap alternatives (1 2, or 3 hops). This pre-tree topology awareness is the first step in our '*low voltage*, or low-reachability architecture. It will sound insane to many network engineers, that there is such a low limit to the number of 'hops' of awareness we begin with, and even more insane to what may appear (artificially) to limit the number of hops to 2 or 3 for this awareness to extend.

Overlapping Liveness Tiles

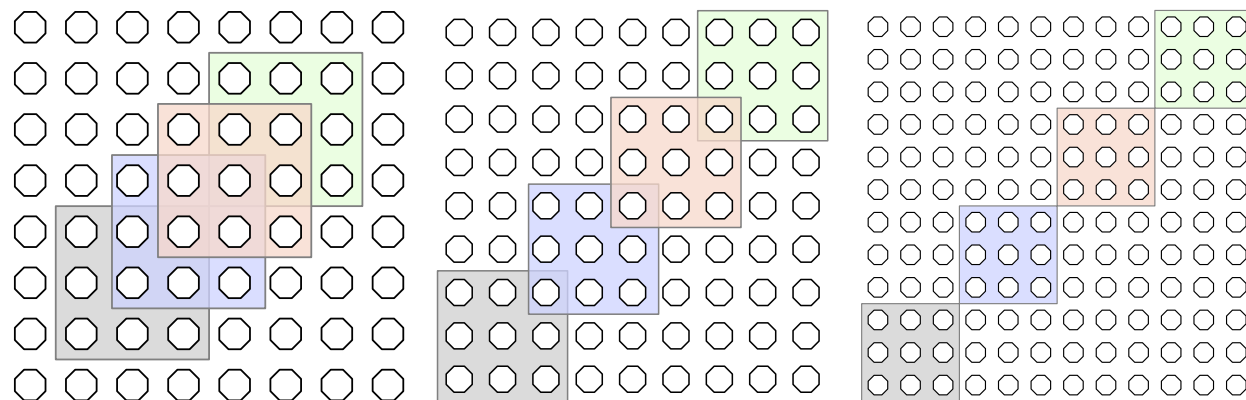


Figure 3: Hypercell Liveness 1 Hop ($3 \times 3 = 9$ cells) 2-Hop ($7 \times 7 = 49$ cells). 3-hop Topology awareness. Link Liveness Awareness = 8, 64 and 512 links (with overlaps from multiple orientations).

¹Except for possible compatibility with protocols required by switches.

²Such as what might be needed for high speed trading and other transaction systems where performance is paramount.

³Directly related to [Wolfram's Multiway Systems](#).

Progressive Knowledge Balance Awareness

Low-voltage⁴ = low reachability.

Spekkens toy Qubits can be in 1 of 4 states. One Qtoken is 4 bits. Two entangled Qtokens have $4 \times 4 = 16$ states. When Carol is the referee between of Alice and Bob, she can decide (select between) the available orderings depending on how her system was ‘prepared’. Carol is a referee. She decides what she sees, but otherwise does not interfere with the ordering of events⁵.

Instead of Carol, the referee, we have Charlie, the Captain of the team who decides what orderings to ‘play’; the Coordinator, who presents the algorithm to coordinate transaction schedules in the substructure.

Charlie is the only one who can ‘see’ all the tokens, and can ‘retain’ as many of them as necessary to coordinate the resource utilization from all the Alice’s and Bob’s.

Charlie is smart enough to write coordination algorithms that do not have failure modes or deadlocks. Charlie can also recruit one or more of his directly-connected neighbors to take over in case his cell fails (voluntarily or involuntarily) or needs to be taken down while a transaction is in progress.

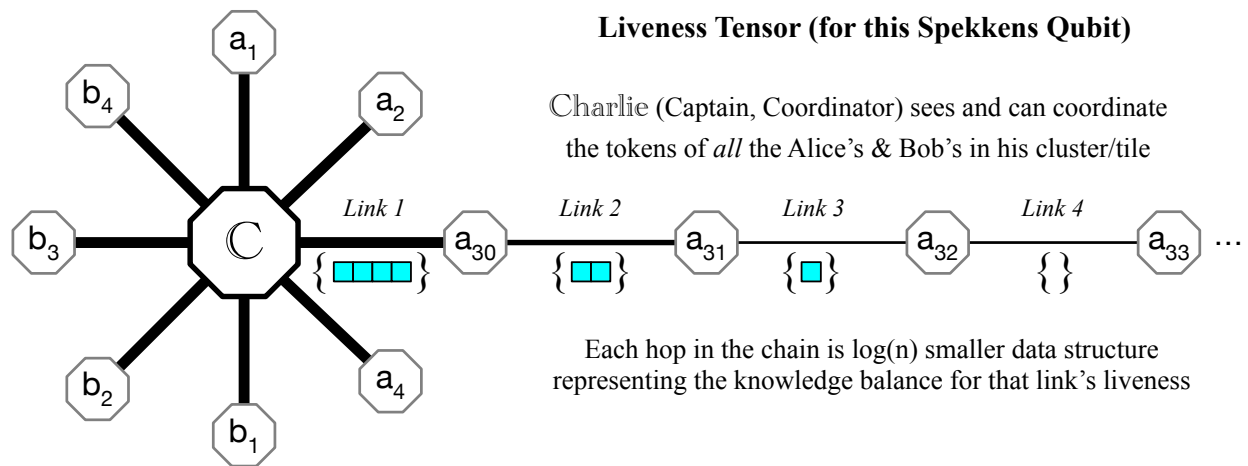


Figure 4: Progressive Digital Decoherence

O Captain! my Captain!

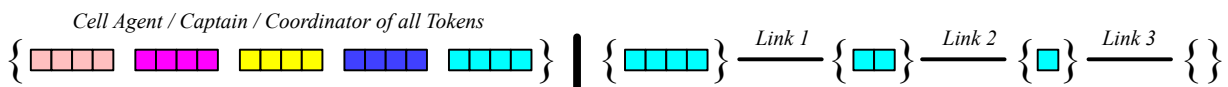


Figure 5: Captain Coordinator

The center cell is the ‘Captain’ of the team, the coordinator. Applications decide the set of compatible (multiway) constraints, but the Captain decides which among them should be selected – rejecting inconsistent ones – and re-arranging orderings which are close, by cooperating with other nodes in the FPGA substructure.

I’m OK, Are You OK?

The **liveness** conversation is a tuple {I’m OK, are you OK?}. A statement about what I know about myself, and a question I am asking my neighbor. The informational states are 0, +1 and –1. We treat information as the answer to a yes/no question. With 0 meaning ‘I don’t know yet’.

While figure ?? describes the distinction between three computational entities (variables) on a single network channel in series, this can be generalized to N variables (computational entities) on a chain – or between pairs of computational entities executing in a single address space on a single multicore computer.

⁴We use high voltage as our metaphor to guide us through the conceptual jungle of what’s wrong with modern networks, and why they are so fragile, error prone, and insecure. High voltage is metaphorically incompatible with human and appliance *safety*.

⁵What Mark Burgess calls [Client Race Adjudication](#).

Figure ?? shows a *chain* of pairs. This can be generalized to a *tree* of pairs. This is the Triangle Demo (see Amin Baumeler in last section).

Every pair of interactions requires four Spekkens Epistemic restriction bits. Triplets (A , B and C) shown in the above example will require twelve Spekkens bits, and a chain of N computational entities in series will require $4(N - 1)$ Spekkens bits for each pairwise interaction. i.e. some combinations of interactions in the connected FPGA logic will enable the selection of a compatible subset of a [multiway execution](#).

The idea is that we can do some number of roundtrip phases in the FPGA substructure without having to go up to the X86 (or other Host) processor, over the slow PCIe bus. By keeping the events (and their indefinite causal order) ‘within’ the FPGA substructure, we hide (or withhold) these events from other parts of the system (they don’t need to know what they don’t need to know). Reserving communication with higher level systems to only when something changes (a surprise), or some process it has requested completes.

Hypercells (self cell+ 8 connected Moore neighborhood)

The hypercell concept (a self-cell with 8 direct connections to it's neighbors), has surprising benefits:

- A self-cell now has (for free) 8 neighbors who will proxy for them in case they fail.
- A self-cell has an immediately available (1 hop) cohort consensus (to interact with the outside world).
- A free 9-cell consensus cluster (itself + 8 others). only 3, 5, or 7 need to reply to [flexible consensus](#).
- Awareness one hop out for dynamic (4-bit Spekkens liveness)
- Awareness 2 hops out for static (2-bit liveness).
- Awareness 3-hops out for topology changes.
- Hypercells can be logical, they can move (see Fig. 6).

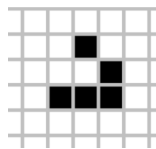


Figure 6: Hypercell Glider

Gliders are [coherent space-time structures](#). From a system redundancy perspective, they carry their own self-repairing redundancy with them as they move.

Category Theory as a Transformer

We use the metaphor of Tesla's multiphase generator invention, where AC enabled transformation to simple, safe, and secure 'low voltage' within buildings, but still allowing the physics and economics of high voltages to span long distances between power stations and substations, and then down to neighborhoods.

In the spirit of Kalman: 'Get the physics right, the rest is mathematics'. We think this goes one stop further: get the physics right, and the economics will follow – just as it did with Tesla's Alternating current, which allowed insulating transformers to safely step down the high voltage of long distance transport of electricity, to make it safe and user-friendly for humans.

We use Compositional Thinking⁶ in Cyberphysical Systems Theory to guide the design of our product.

We use notions of category theory, and functors – structure preserving 'maps' between categories that can translate a the semantics of the application domain to the semantics of the infrastructure domain.

"The most general notion of a category is that of an algebraic structure on a graph. A category contains a collection of objects and a collection of arrows (including an identity arrow) which can formally be joined by a composition rule. Examples of useful categories include the category of sets and functions, of graphs and graph trans- formations, and of states and transitions between them.

The notion of functoriality between syntax and semantics we talked about previously is implemented through a structure preserving map between categories. Functors are a predefined way of transforming some data and their associated interconnections to another form. For example, a functor can be taking the powerset of a set, where we have predefined operations on how to do that between the same category, namely the category of sets."

"We can think of a functor in the application of cyber-physical system design as preserving semantics across the currently distinct algebras that define different interpretations of requirements, system behaviors, and system ar- chitectures, meaning that they assign a particular algebra to a specific model view.

Often we would like to model operations happening in parallel when speaking about cyber- physical systems. The constrained definition of a category is capable of modeling sequential processes but not parallel ones. A particularly useful type of category in system science is one that is monoidal. Monoidal categories

⁶Georgios Bakirtzis, Eswaran Subrahmanian, Cody H. Fleming: [Compositional Thinking in Cyberphysical Systems Theory](#).

extend this definition precisely to equip with a tensor. product functor, which models processes and/or operations that need to happen in parallel.”

Fortunately that’s exactly what we have with our Liveness Tensor. By creating a new (but Ethernet compatible) layer 2 protocol we can map from the high voltage of A2A (Any to Any) IP addressing to a new form of distance-constrained ‘relative addressing’ which eliminates many of the challenges inherent in the (also A2A) form of existing Ethernet protocols.

”By having the mathematical structure of a category it is now possible to translate between model views as long as they form categories and their behavior can be represented as an algebra. This is achieved by defining a precise way, that can also be implemented algorithmically, of trans- lating the objects from one category to another”

“The promise of category theory is not that it will change the mathematics of a particular application field but, rather, that it will organize the information such that there might be new insights and techniques associated with trans- lating between data (verified composition) and managing complexity (scalability).”

“Compositionality is one of the open problems in cyber-physical system design, as is argued in the NIST cyber-physical system framework [5, pages 11-12].” “The main benefits of applied compositional thinking is verified composition and scalability of different system models.” “algebras, even implicitly, exist at any given level of system design and their composition gives rise to well- defined traces in models.”

“One way to partially combat the increasing but necessary complexity of those systems we need to make sure that all these algebras are traceable. By being traceable we are therefore able to apply several lightweight formal methods within one overarching model.”

Product

We are a software infrastructure company. We write software. Fortunately, these days, that also includes the translation of that software into functions that reside in FPGA’s on (otherwise off-the-shelf) SmartNICs.

Product Requirements that Solve a Real Problem

1. Low-voltage Local (short range) Low valency connectivity before High voltage, sparse, long-distance
2. Non-clonable Functions, unique to the hardware.

Whereas IP, and Ethernet too for that matter, hides the wiring diagram of the house or datacenter, we explicitly create and manage it, using a Graph Virtual Machine (GVM). We use standard Ethernet ‘iron pipes’ without the switches which cause all the problems. Direct connections (which can use inexpensive short-range copper cables). The economics is astounding. Where conventional networks are dominated by the cost of cabling (and modules which translate to optical), our networks have low insertion loss, short range, and are private. We can also eliminate the eavesdropping and other ‘information leakage’ issues with insecure switches and routers. And yet, we can scale locally within a rack to 10’s to 100’s nodes, and with a relative (tree-based) addressing structure, we can become ‘scale independent’.

“An example of a functorial operation between categories of sets is transforming a set to its power set. A monoidal functor comes with comparison morphisms that involve the tensor product, allowing us to use the same idea in the setting of parallel operations. This might seem like new and confusing jargon for those unfamiliar with category theory but once it is understood it allows us to summarize a vast number of mathematical techniques as well as formal methods and system models by allowing us to assign algebras that contain parallel processes.”

Wiring diagrams before Algebras

“The general principle of applied compositional thinking is the following.

1. Wiring diagrams define how things ought to be connected and in what way.
2. Algebras assign a particular formalism or behavior to the empty processes inhabiting the wiring diagram.

Wiring diagrams are the theory of the class of systems we want to model but only insofar as they are assigned a behavioral model through an algebra (monoidal functor).”

Algebras before Code

This magic doesn't require expensive development (everything we need to get started is off the shelf). Over time, the economics of mass production can be used to dominate the market with 'hardware enclaves' that are not only far more resilient and secure, they are much lower latency and capable of utilizing bandwidth far more effectively.

Mathematics is the Poetry of Logic. Category Theory helps remove the ad-hoc human judgement of architects as artists and replaces them with the discipline of getting out of one's comfort zone and replacing it with the discipline of physics models and mathematical tools.

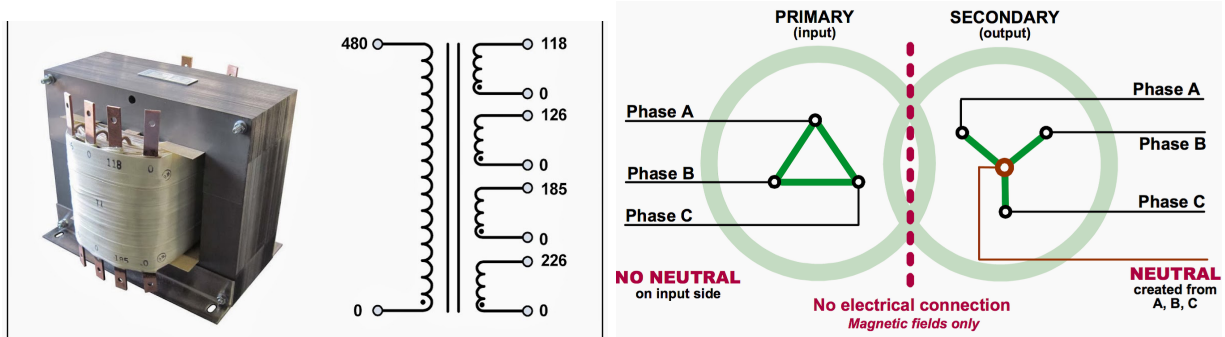


Figure 7: Transformer - this is what we do for network protocols – in software

With apologies to Rudy Kalman and Jan Williams⁷

- Get the physics right.
- The rest is mathematics
- (Added by me) The economics will follow

“An interconnected system is a system that consists of interacting subsystems modeled by tearing, zooming, and linking.”

The Physics of Cabling

Short range (low voltage), copper wins. Long range (High voltage) Fiber wins. It's not just cost, it's physics. [Fiber vs. Copper](#). To make this distinction clear. We can identify our 'low-voltage' world with low-cost, low reachability *L2'* protocols, and the outside 'high-voltage' world with fiber.

Inevitable Architectural Shift

Another way to look at this shift is

Promise Theory

An effective way to specify requirements of the system behavior is to define them in Promise Theory: a model of *cooperation* rather than *impositions*. It allows each of the elements to define what they feel they are capable of doing under the circumstances, which includes the resources available to them, and to try to maintain best-effort while not incurring the increasing brittleness and cascading failure modes of command and control style infrastructure.

[Is Promise Theory is a category?](#) “Promise Theory (PT) and Category Theory (CT) occupy different layers of process modeling. In principle, CT is about mappings and relationships, whereas Promise Theory is a description of causal intent. CT has no ‘locations’ or ‘agents’ that cause things to happen, so all arrows in CT look like impositions from a PT perspective. However, PT does little to formalize the promise bodies it refers to, as its focus is often on causal relationships, or interactions rather than the specific semantics of the

⁷Jan C. Willems: [The Behavioral Approach to Open and Interconnected Systems](#).

interactions. This is where CT goes into ultimate detail. So, one way to think about it is that PT defines the spacetime arena for processes, while CT is a layer of detail, which explicates and details the nature of the promises, on top of PT layer.”

Security Tensor

“For a restriction in each subsystem in the composite we can represent that restriction as a set of allowable pairs that can be observed as inputs and outputs and from this formulation we will always be able to calculate the allowable input/output pairs for the composite system in terms of the specific wiring given by the architecture of the wiring diagram.”